PATHOLOGY · MCQ SET

Glomerular Disease — Nephrotic & Nephritic Syndromes

Pathology GUE Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty. Attempt every question before checking the answer key on the last page.

4 Easy **10 Medium** **6 Hard**

1. What is the most common cause of nephrotic syndrome in children? EASY

A. Membranous glomerulopathy
B. Minimal change disease
C. FSGS
D. MPGN
E. IgA nephropathy

2. What is the most common cause of nephrotic syndrome in adults? EASY

A. Minimal change disease
B. Membranous glomerulopathy
C. Diabetic nephropathy
D. Amyloidosis
E. Lupus nephritis

3. Which stain demonstrates amyloid as red under light microscopy and apple-green birefringence under polarized light? EASY

A. PAS stain
B. Congo red stain
C. Silver stain
D. Masson trichrome
E. H&E stain

4. Which syndrome is characterized by hematuria, subnephrotic proteinuria, oliguria, azotemia, and hypertension? EASY

A. Nephrotic syndrome
B. Nephritic syndrome
C. Acute tubular necrosis
D. Chronic kidney disease
E. Renal tubular acidosis

5. A 4-year-old develops nephrotic-range proteinuria after a viral URI. Biopsy shows normal glomeruli on light microscopy and no immune deposits on immunofluorescence. EM shows diffuse foot process effacement. Which best describes the expected treatment response? MEDIUM CASE

A. No response to any therapy
B. Dramatic response to corticosteroids
C. Requires dialysis
D. Responds only to plasmapheresis
E. Spontaneously worsens regardless of treatment

6. Which EM finding is characteristic of membranous glomerulopathy? MEDIUM

A. Subepithelial deposits separated by GBM spikes
B. Subepithelial 'humps'
C. Mesangial IgA deposits
D. Basket-weave GBM
E. Wire-loop lesions

7. A renal biopsy shows glomeruli that are large, lobular, and hypercellular, with a 'tram-track' or double-contour appearance of the capillary wall. What is the diagnosis? MEDIUM

A. Minimal change disease
B. Membranous glomerulopathy
C. FSGS
D. Membranoproliferative GN
E. Post-streptococcal GN

8. A diabetic patient's renal biopsy shows ovoid, laminated PAS-positive nodules within the mesangial core, surrounded by dilated capillary loops. This is best termed: MEDIUM CASE

A. Wire-loop lesion
B. Nodular glomerulosclerosis (Kimmelstiel-Wilson)
C. Crescent formation
D. Tram-track appearance
E. Basket-weave pattern

9. A patient with multiple myeloma develops nephrotic syndrome. Renal biopsy shows amyloid deposits composed of monoclonal light chains. This is classified as: MEDIUM

A. AA amyloidosis
B. AL amyloidosis
C. Transthyretin amyloidosis
D. Dialysis-associated amyloidosis
E. Senile systemic amyloidosis

10. A lupus patient's renal biopsy shows diffuse proliferative glomerulonephritis with wire-loop lesions resulting from subendothelial immune complex deposition. Which WHO lupus nephritis class is this? MEDIUM CASE

A. Class I
B. Class II
C. Class III
D. Class IV
E. Class VI

11. A 7-year-old develops hematuria, edema, and hypertension two weeks after an episode of impetigo. What is the most likely diagnosis? MEDIUM

A. Minimal change disease
B. Membranous glomerulopathy
C. Post-streptococcal glomerulonephritis
D. IgA nephropathy
E. Alport syndrome

12. Which electron microscopic finding is pathognomonic for post-streptococcal glomerulonephritis? MEDIUM

A. Subepithelial 'hump'
B. Subepithelial spikes
C. Mesangial deposits only
D. Basket-weave GBM
E. Linear GBM staining

13. A patient presents with rapidly progressive renal failure and pulmonary hemorrhage. Immunofluorescence shows linear staining for IgG along the GBM. Which type of RPGN and which syndrome does this represent? MEDIUM CASE

A. Type I RPGN — Goodpasture syndrome
B. Type II RPGN — postinfectious GN
C. Type III RPGN — pauci-immune (ANCA)
D. Type II RPGN — IgA nephropathy
E. Type III RPGN — Alport syndrome

14. Crescents in rapidly progressive glomerulonephritis form from proliferation of which cell type, together with fibrin and leukocytes? MEDIUM

A. Podocytes
B. Mesangial cells
C. Parietal epithelial cells
D. Endothelial cells
E. Tubular epithelial cells

15. A young man has hematuria, sensorineural hearing loss, and lens dislocation; his mother has isolated hematuria. Renal biopsy EM shows a basket-weave GBM pattern. Which gene/protein defect explains this X-linked inheritance pattern? HARD CASE

A. WT1
B. Alpha-5 chain of type IV collagen
C. Podocin (NPHS2)
D. PKD1
E. Nephrin

16. Which statement correctly distinguishes IgA nephropathy from post-streptococcal GN? HARD

A. IgA nephropathy requires immunofluorescence for diagnosis and often follows mucosal infections by only 1-2 days; PSGN follows strep infection by 1-4 weeks with granular deposits
B. Both are diagnosed by light microscopy alone
C. PSGN is the most common GN worldwide; IgA nephropathy is rare
D. IgA nephropathy shows subepithelial humps; PSGN shows mesangial IgA deposits
E. Neither shows immune complex deposition

17. Regarding RPGN immunofluorescence patterns, which pairing is correct? HARD

A. Type I — granular; Type II — linear; Type III — scanty
B. Type I — linear; Type II — granular; Type III — scanty/pauci-immune
C. Type I — scanty; Type II — linear; Type III — granular
D. All three types show identical linear staining
E. All three types show identical granular staining

18. A patient with chronic hepatitis C develops nephrotic-range proteinuria with hematuria. Biopsy shows hypercellular, lobular glomeruli with GBM duplication. Which primary glomerular disease does this represent as a secondary phenomenon, and which complement pathway is activated in the immune-complex-mediated form? HARD CASE

A. Membranous glomerulopathy; alternative pathway
B. MPGN; classic complement pathway
C. Minimal change disease; no complement activation
D. FSGS; lectin pathway
E. Post-streptococcal GN; alternative pathway

19. Which of the following correctly matches the amyloid type with its clinical association? HARD

A. AL amyloid — serum amyloid A, chronic inflammation
B. AA amyloid — monoclonal light chains, plasma cell dyscrasia
C. AL amyloid — monoclonal light chains, plasma cell dyscrasia (e.g. myeloma)
D. AA amyloid — monoclonal light chains, myeloma
E. Both AL and AA arise exclusively from light chains

20. A patient has nonselective proteinuria, subepithelial deposits with GBM spikes on EM, and poor response to corticosteroids. A second patient has selective (albumin) proteinuria, no deposits on IF, isolated foot process effacement, and dramatic steroid response. Which diagnoses, respectively, best fit these two patients? HARD CASE

A. Membranous glomerulopathy; minimal change disease
B. Minimal change disease; membranous glomerulopathy
C. FSGS; MPGN
D. MPGN; FSGS
E. Diabetic nephropathy; amyloidosis

A Answer key & rationale

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	**B**	Minimal change disease is the most common cause of nephrotic syndrome in children.
2	**B**	Membranous glomerulopathy is the most common cause of nephrotic syndrome in adults.
3	**B**	Congo red stain gives red color under light microscopy and apple-green birefringence under polarized light for amyloid.
4	**B**	Nephritic syndrome presents with hematuria, subnephrotic proteinuria, oliguria, azotemia, and hypertension.
5	**B**	Minimal change disease shows a dramatic response to corticosteroid therapy.
6	**A**	Membranous glomerulopathy shows subepithelial deposits separated by small spikes in the GBM matrix on EM.
7	**D**	Large, lobular, hypercellular glomeruli with a tram-track/double-contour appearance define membranoproliferative GN.
8	**B**	Ovoid, laminated PAS-positive nodules within the mesangial core define nodular glomerulosclerosis (Kimmelstiel-Wilson), seen in diabetic nephropathy.
9	**B**	AL amyloidosis consists of monoclonal light chains and occurs in patients with plasma cell dyscrasias such as multiple myeloma.
10	**D**	Diffuse proliferative lupus nephritis with wire-loop lesions is WHO Class IV, the most severe class.
11	**C**	Hematuria, edema, and hypertension 1-4 weeks after streptococcal pharyngitis or impetigo describe post-streptococcal GN.
12	**A**	A subepithelial electron-dense 'hump' is the pathognomonic EM finding of post-streptococcal GN.
13	**A**	Linear IgG staining along the GBM with pulmonary hemorrhage and rapidly progressive renal failure describes Type I RPGN / Goodpasture syndrome.
14	**C**	Crescents form from proliferation of parietal epithelial cells, together with fibrin deposition and leukocyte infiltration.
15	**B**	Alport syndrome results from mutation in the gene for the alpha-5 chain of type IV collagen, producing a basket-weave GBM; the X-linked form is most common.
16	**A**	IgA nephropathy is diagnosed only by immunofluorescence and classically follows mucosal infection by 1-2 days ('synpharyngitic'); PSGN follows strep infection by 1-4 weeks with granular immune deposits.
17	**B**	RPGN Type I shows linear IF staining (anti-GBM), Type II shows granular staining (immune complex), and Type III shows scanty/ pauci-immune staining (ANCA-associated).
18	**B**	Secondary MPGN can arise from hepatitis C infection; immune-complex-mediated MPGN activates the classic complement pathway.
19	**C**	AL amyloid is composed of monoclonal light chains and occurs in plasma cell dyscrasias (e.g. myeloma); AA amyloid is derived from serum amyloid A in chronic inflammation.
20	**A**	Nonselective proteinuria with subepithelial deposits/spikes and poor steroid response describes membranous glomerulopathy; selective proteinuria with isolated foot process effacement and dramatic steroid response describes minimal change disease.